

Motility Training App

Aneequa Sundus

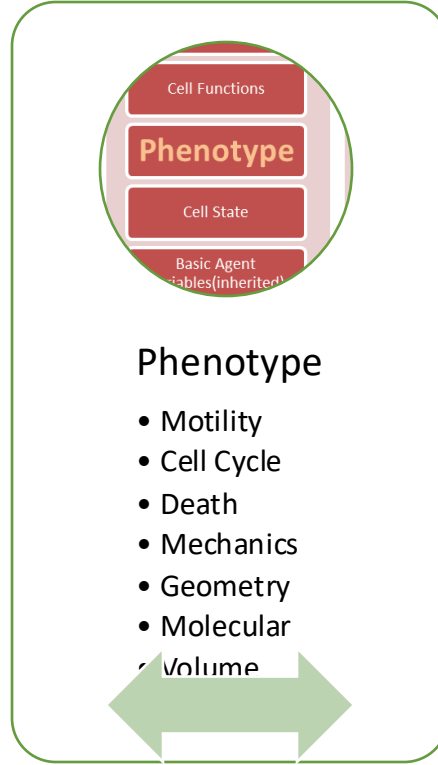


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Context



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Context

Microenvironment

Domain Size

Voxel Size

Substrates

Boundary Conditions

Cells

Custom Data

Cell Functions

Phenotype

Cell State

Basic Agent
Variables(inherited)

Cell Definition

Global Variables

User Parameters

PhysiCell Constants

List of cells

SVG options

MultiCellIDS options

Default Cell Definition

Default microenvironment



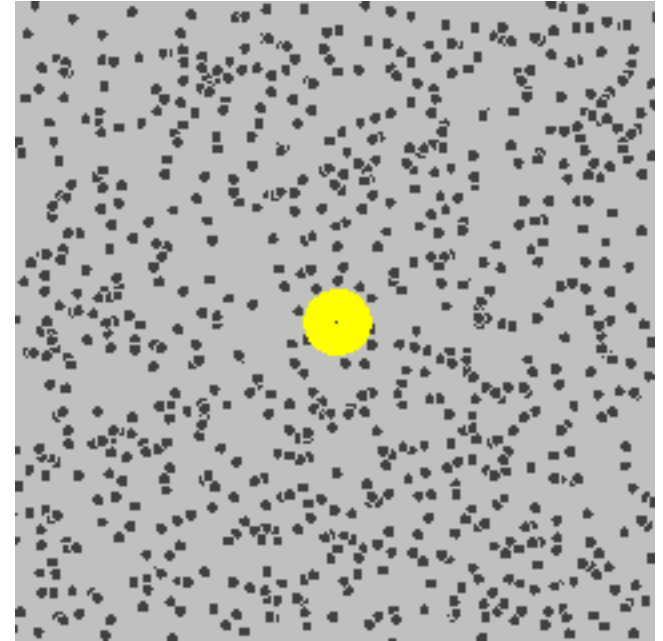
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Background

- Cells move through motility.
- Mathematically modeled as a biased Random walk.
- No bias : Brownian Motion
 - ✦ Cells move randomly due to their collisions
 - ◆ Yellow Particle
- With bias : Random Biased Motion
 - ✦ Cells select movement directions according to some rules
 - ◆ e.g Chemotaxis



“Brownian Motion and Random Walks.” MIT,
web.mit.edu/8.334/www/grades/projects/projects17/OscarMickelin/brownian.html.



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Mathematics: Variables and Definitions

- **Migration speed:** Migration speed is the speed at which the cells move in $\mu\text{m}/\text{min}$.
- **Migration bias direction:** Preferred migration direction
- **Migration bias:** Impacts the direction that a cell moves.
 - ✦0 ☐ Random Walk
 - ✦1 ☐ Deterministic Motion along migration bias direction
- **(Mean) persistence time:** The mean duration in minutes that a cell moves in its current direction before choosing a new motility vector to follow.



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Mathematics: Models

- How is motility direction chosen and determines the bias direction?

$$\mathbf{d}_{\text{mot}} = \frac{b \mathbf{d}_{\text{bias}} + (1 - b)\boldsymbol{\xi}}{\|b \mathbf{d}_{\text{bias}} + (1 - b)\boldsymbol{\xi}\|}$$

Where;

$\mathbf{d}_{\text{mot}}$: direction of the motility

b : the level of bias in motility (migration bias)

$\boldsymbol{\xi}$: random unit vector of length 1,

$\mathbf{d}_{\text{bias}}$: direction of the bias

- How is the motility velocity computed?

$$\mathbf{v}_{\text{mot}} = s_{\text{mot}} \mathbf{d}_{\text{mot}}$$

Where;

s_{mot} : migration speed

- How often do cells change motility direction?

Persistence Time



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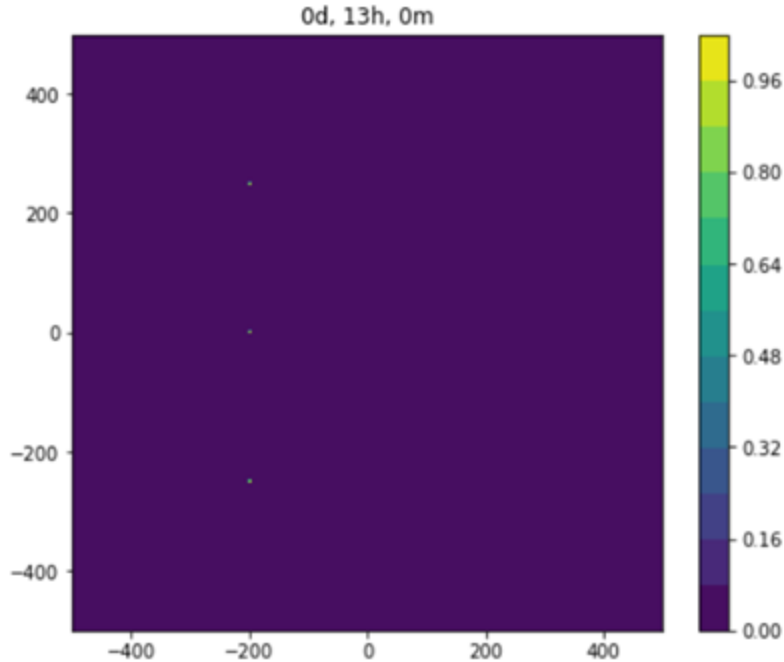
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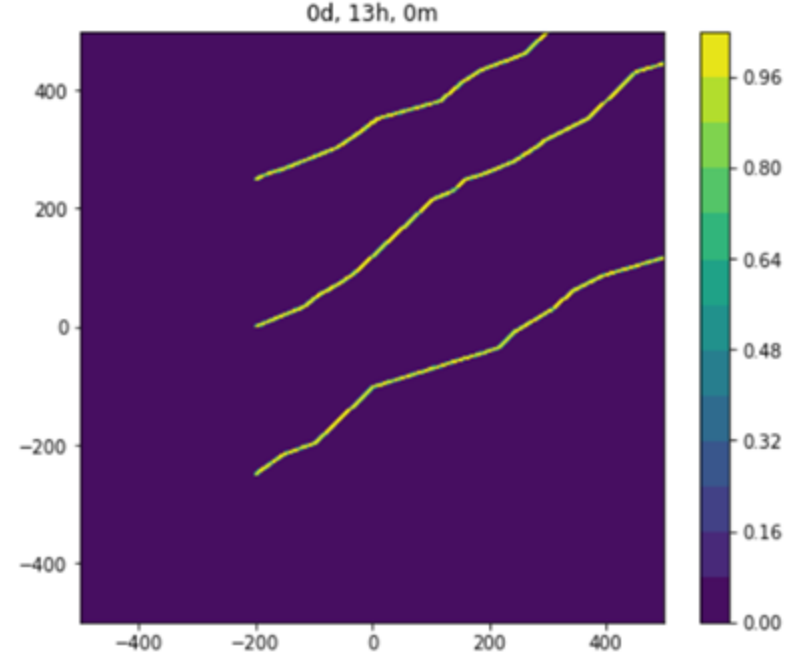
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Expected Results (is_motile)



is_motile = False



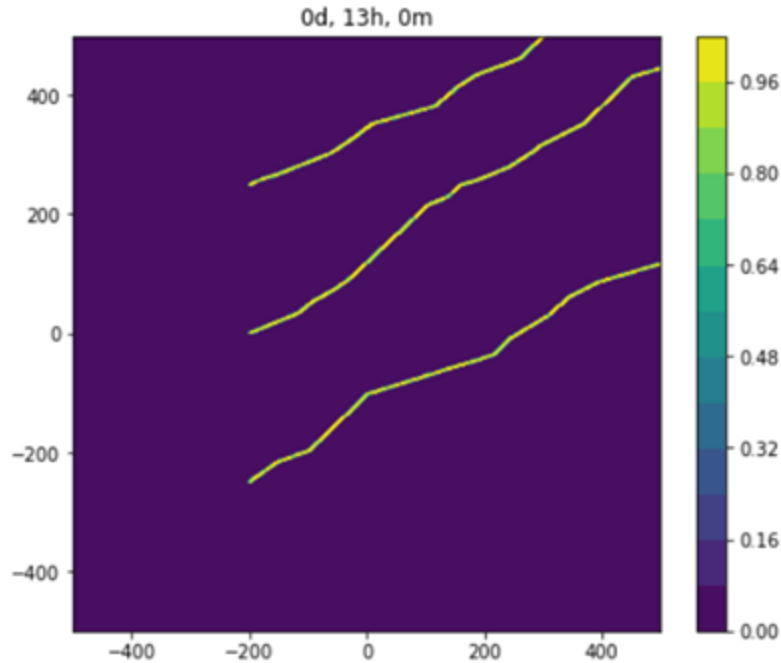
is_motile = True



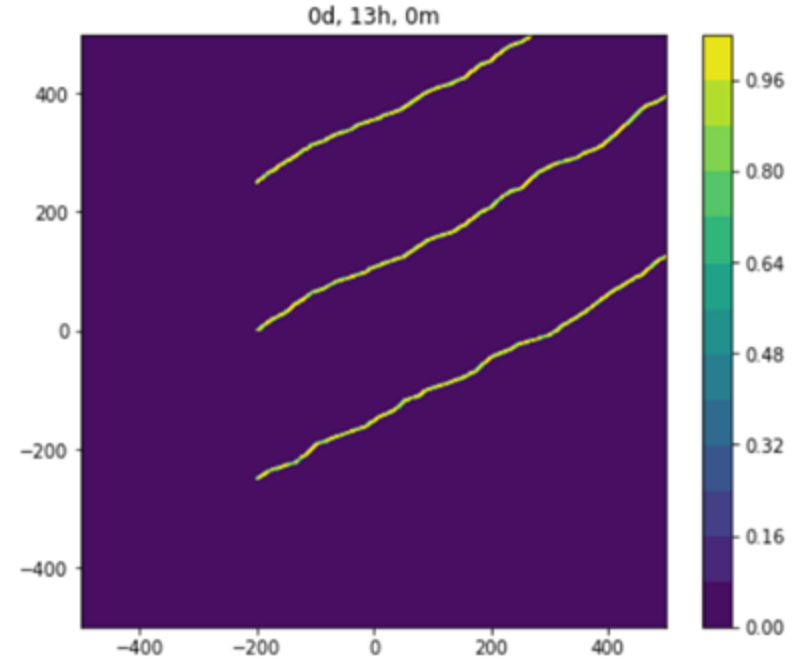
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Expected Results (Persistence_time)



persistence_time = 30 min



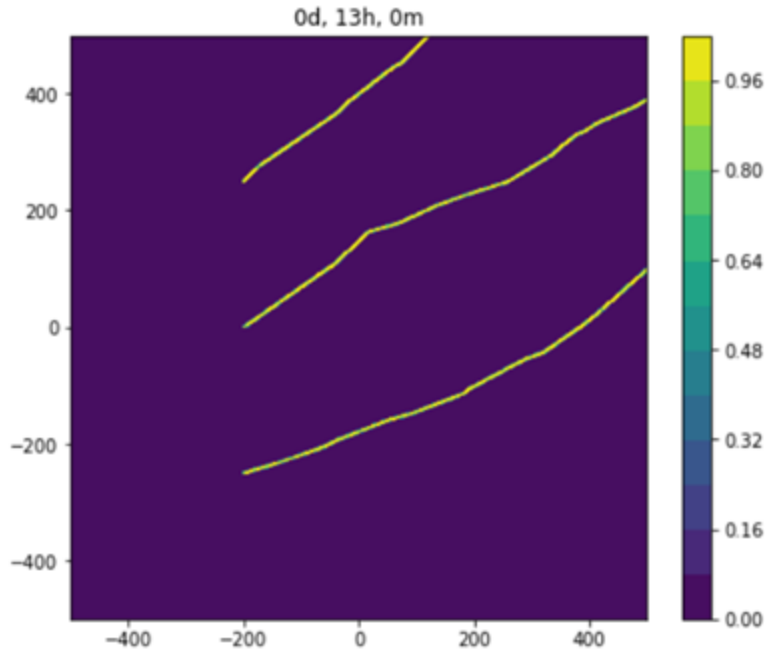
persistence_time = 5 min



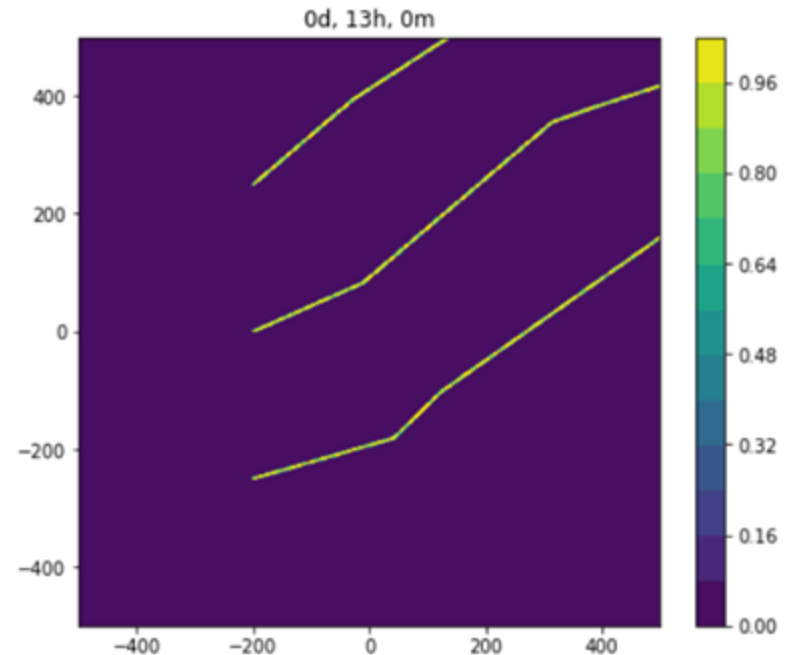
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Expected Results (migration_speed)



Migration_speed = 2 micron/min⁻¹



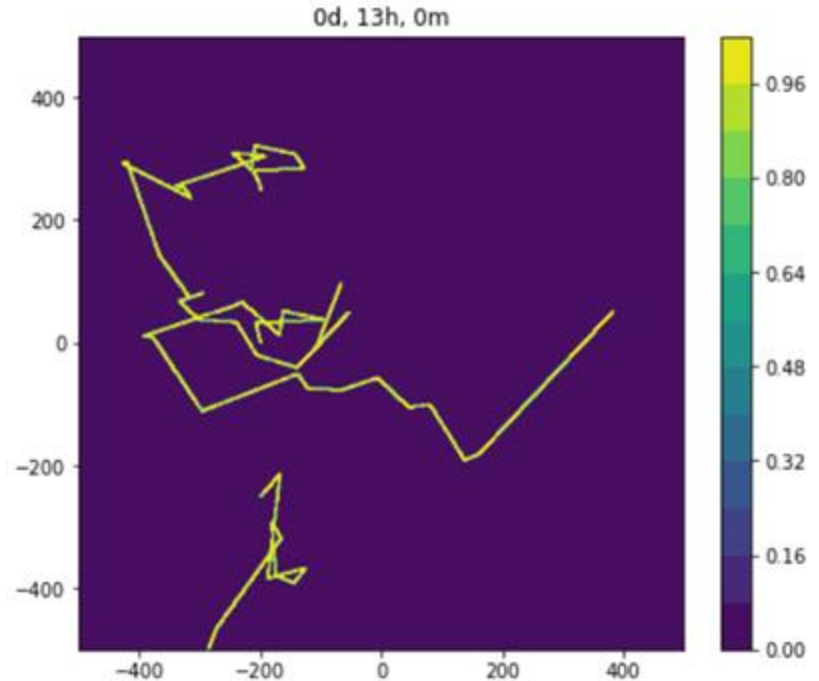
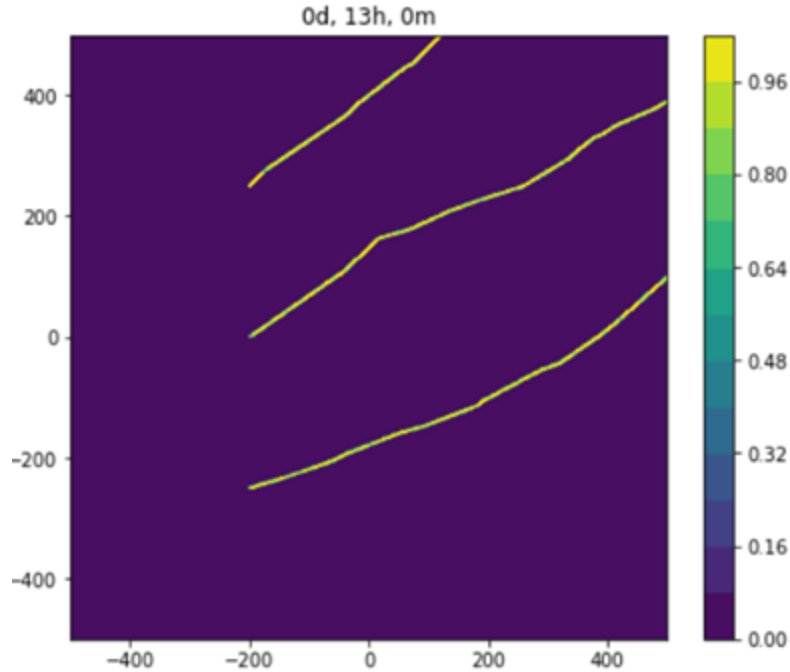
Migration_speed = 10 micron/min⁻¹



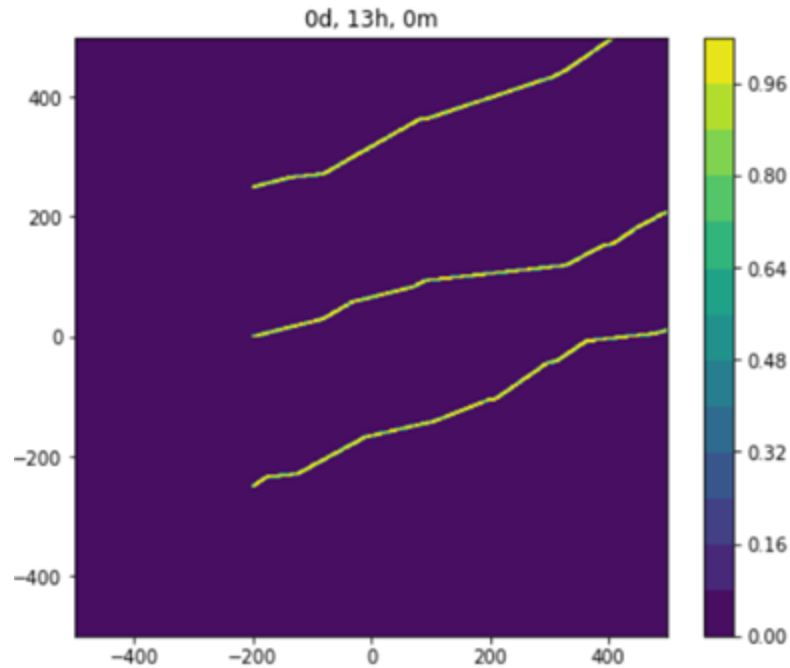
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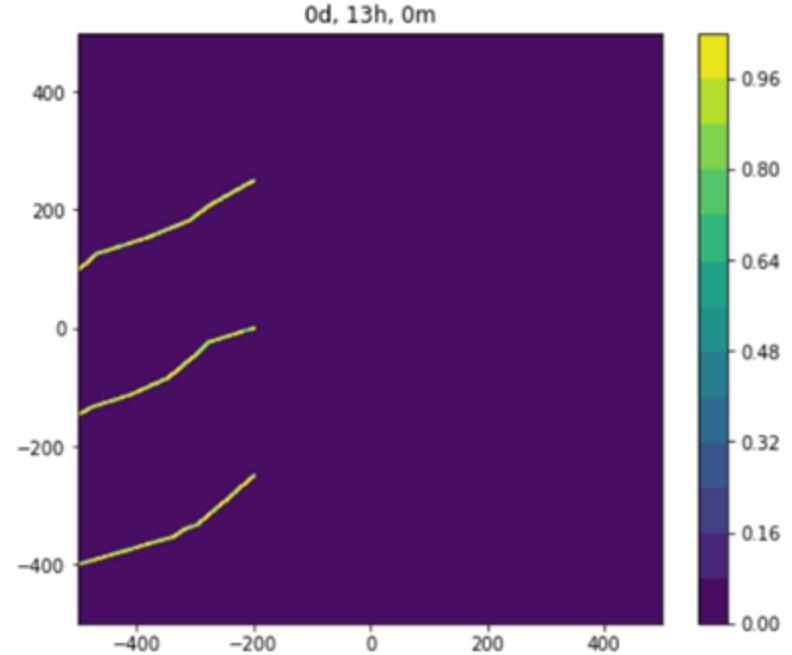
Expected Results (migration_bias)



Expected Results (bias_migration_angle)



Migration_bias_angle = 20



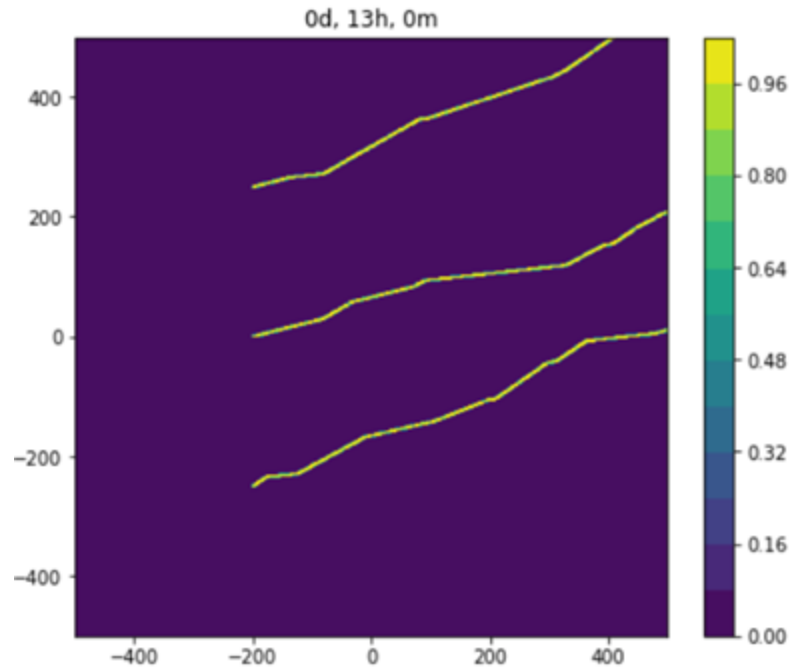
Migration_bias_angle = 210



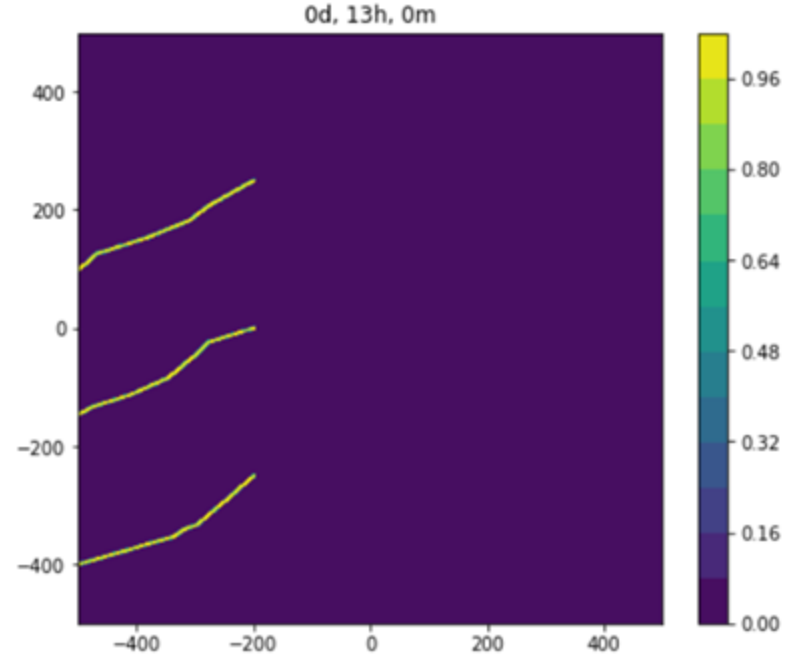
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Expected Results (bias_migration_angle)



Migration_bias_angle = 20



Migration_bias_angle = 210



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Class structure: Data

Key parameters in phenotype.motility

- `bool is_motile;`
 - Set to true if cell is / can actively migrate
- `double persistence_time;`
 - mean time to keep going in one direction before resampling for a new direction.
- `double migration_speed;`
 - migration speed along chosen direction, in absence of all other adhesive / repulsive forces
- `std::vector<double> migration_bias_direction;`
 - random motility is biased in this direction (e.g., chemotaxis)
- `double migration_bias;`
 - how biased is motility
 - if 0, completely random. if 1, deterministic along the bias vector
- `bool restrict_to_2D;`
 - Set to true if you want to guarantee 2D motion
- `std::vector<double> motility_vector;`
 - direction of migration



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Example

custom.cpp

```
Cell* pCell= create_cell();  
pCell->phenotype.motility.is_motile= true;  
pCell->phenotype.motility.migration_speed= 1.0;  
pCell->phenotype.motility.migration.persistence_time= 20;  
pCell->phenotype.motility.restrict_to_2D = true;  
pCell->phenotype.motility.migration_bias= 0.8;  
std::vector<double> temp = { 0.1 , 0.2, 0.0 };  
normalize( temp );// makes it's length 1  
pCell->phenotype.motility.migration_bias_direction= temp;
```



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Example

custom.cpp

```
Cell* pCell= create_cell();
pCell->phenotype.motility.is_motile= parameters.bools( "is_motile" );
pCell->phenotype.motility.migration_speed= parameters.doubles( "migration_speed" );
pCell->phenotype.motility.migration.persistence_time= parameters.doubles("persistence_time");
pCell->phenotype.motility.restrict_to_2D = true;
pCell->phenotype.motility.migration_bias= parameters.doubles( "migration_bias" );
std::vector<double> temp = { 0.1 , 0.2, 0.0 };
normalize( temp );// makes it's length 1
pCell->phenotype.motility.migration_bias_direction= temp;
```

PhysiCell_settings.xml

```
<is_motile type="bool" units="none" description="true if cells are motile">True</is_motile>
<migration_speed type="double" units="micron/min" description="migration speed">1.0</
migration_speed>
<persistence_time type="double" units="min" description="mean persistence time">1.0</
persistence_time>
<migration_bias type="double" units="none" description="migration bias parameter">0.8</
migration_bias>
```



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Credits

Lesson Planning: Paul Macklin

Slides: Aneequa Sundus, Kali Konstantinopoulos*, Mary Chen*, Furkan Kurtoglu

Funding:

PhysiCell Development:

- Breast Cancer Research Foundation
- Jayne Koskinas Ted Giovanis Foundation for Health and Policy
- National Cancer Institute (U01CA232137)
- National Science Foundation (1720625)



JAYNE KOSKINAS
TED GIOVANIS

Foundation for
Health and Policy



NATIONAL
CANCER
INSTITUTE



BCRF
BREAST
CANCER
RESEARCH
FOUNDATION

Training materials:

- * Administrative supplement to NCI U01CA232137 (Year 2)



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